ELG 3125 - Signals & Systems

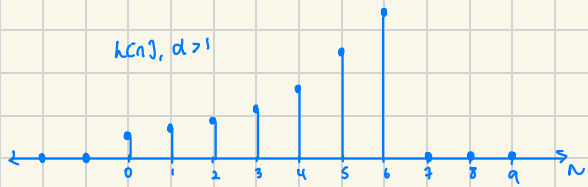
Lecture 4 - Part 2

Convolution Example

Compute the response of an LTI system described by its impulse response, $\alpha > 1$

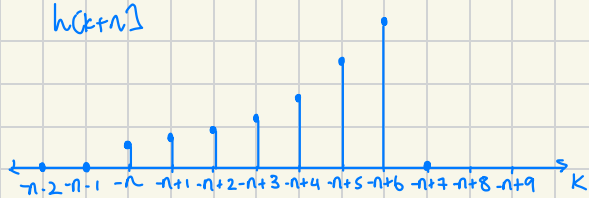
$$h[n] = \begin{cases} \alpha^n, & 0 \le n \le 6 \\ 0, & \text{otherwise} \end{cases}$$ to the input signal $$x[n] = \begin{cases} 1, & 0 \le n \le 4 \\ 0, & \text{otherwise} \end{cases}$$





STEP 1: Draw $x[k]$ and $h[n-k]$, $h[n-k] = h[-k+n]$

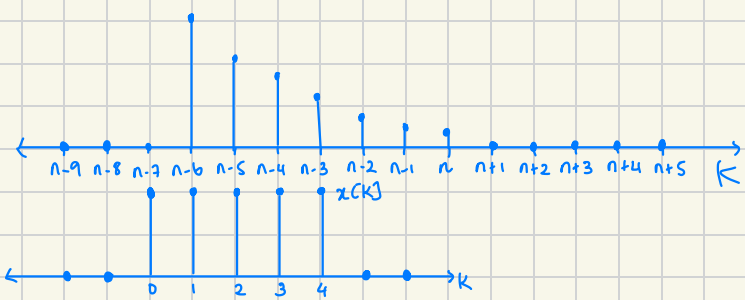




$h[-k+n] = h[n-k]$

$$h[n] = \begin{cases} \alpha^n, & 0 \le n \le 6 \\ 0, & \text{otherwise} \end{cases} \qquad x[k]h[n-k] = \begin{cases} \alpha^{n-k}, & 0 \le k \le n \\ 0, & \text{otherwise} \end{cases}$$

$n - 6 \le 0 \Rightarrow n \le 6$



$n < 0$: no overlap $\Rightarrow$ $\boxed{y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = 0}$

$0 \le n \le 4$: overlap $\Rightarrow$ $y[n] = \sum_{k=0}^{n} \underbrace{x[k]}_{1} \underbrace{h[n-k]}_{\alpha^{n-k}} = \sum_{k=0}^{n} \alpha^{n-k} = \sum_{k=0}^{n} \alpha^n \alpha^{-k} = \alpha^n \sum_{k=0}^{n} \alpha^{-k} = \alpha^n \left(\frac{1-\alpha^{-n-1}}{1-\alpha^{-1}} \right)$

$$\Rightarrow \boxed{y[n] = \alpha^n \left(\frac{1-\alpha^{-n-1}}{1-\alpha^{-1}} \right), \text{ for } 0 \le n \le 4}$$

$n > 4$ & $n - 6 \le 0 \Rightarrow n \le 6 \Rightarrow$ for: $4 < n \le 6$, $x[n]h[n-k] = \begin{cases} \alpha^{n-k}, & 0 \le k \le 4 \text{ (overlap region)} \\ 0, & \text{otherwise} \end{cases}$

$$\Rightarrow x[k]h[n-k] = \sum_{k=0}^{4} \alpha^{n-k} = \sum_{k=0}^{4} \alpha^n \alpha^{-k} = \alpha^n \sum_{k=0}^{4} \alpha^{-k} = \alpha^n \sum_{k=0}^{4} (\alpha^{-1})^k = \alpha^n \left[\frac{1-(\alpha^{-1})^5}{1-\alpha^{-1}} \right] = \alpha^n \left[\frac{1-\alpha^{-5}}{1-\alpha^{-1}} \right] = \frac{\alpha^{n+1} - \alpha^{n-4}}{\alpha - 1} = \frac{-(\alpha^{n-4} - \alpha^{n+1})}{\alpha - 1} = \frac{\alpha^{n-4} - \alpha^{n+1}}{\alpha - 1}$$

→ $y[n] = x[k]*h[n-k] = \frac{\alpha^{n-4} - \alpha^{n+1}}{\alpha - 1}$, for $4 < n \leq 6$

for $n-6 > 4 \Rightarrow n > 4+6 \Rightarrow n > 10$, no overlap between $x[k]$ & $h[n-k]$

→ $y[n] = x[k]*h[n-k] = 0$, for $n > 10$

$n > 6$ & $n - 6 \leq 4 \Rightarrow n \leq 10 \Rightarrow 6 < n \leq 10$

$y[n] = \sum_{k=n-6}^{4} \alpha^{n-k}$, let $r = k - n + 6 \Rightarrow k = r + n - 6$ ⇒ $r + \cancel{n} - 6 = \cancel{n} - 6 \Rightarrow$ $r = 0$

when $k = 4 \Rightarrow r = 4 - n + 6 = 10 - n$

$= \sum_{r=0}^{10-n} \alpha^{n-(r+n-6)} = \sum_{r=0}^{10-n} \alpha^{n-r-\cancel{n}+6} = \sum_{r=0}^{10-n} \alpha^{6-r} = \alpha^6 \underbrace{\sum_{r=0}^{10-n} \alpha^{-r}}_{\text{solve this middle}} = \frac{\alpha^{n+4} - \alpha^7}{1-\alpha}$

→ $y[n] = \frac{\alpha^{n-4} - \alpha^7}{1-\alpha}$, for $6 < n \leq 10$

Overall result :

$$y[n] = \begin{cases} 0, & n < 0 \\ \frac{1-\alpha^{n+1}}{1-\alpha}, & 0 \leq n \leq 4 \\ \frac{\alpha^{n-4} - \alpha^{n+1}}{\alpha - 1}, & 4 < n \leq 6 \\ \frac{\alpha^{n-4} - \alpha^7}{1-\alpha}, & 6 < n \leq 10 \\ 0, & n > 10 \end{cases}$$

Overall plot :

